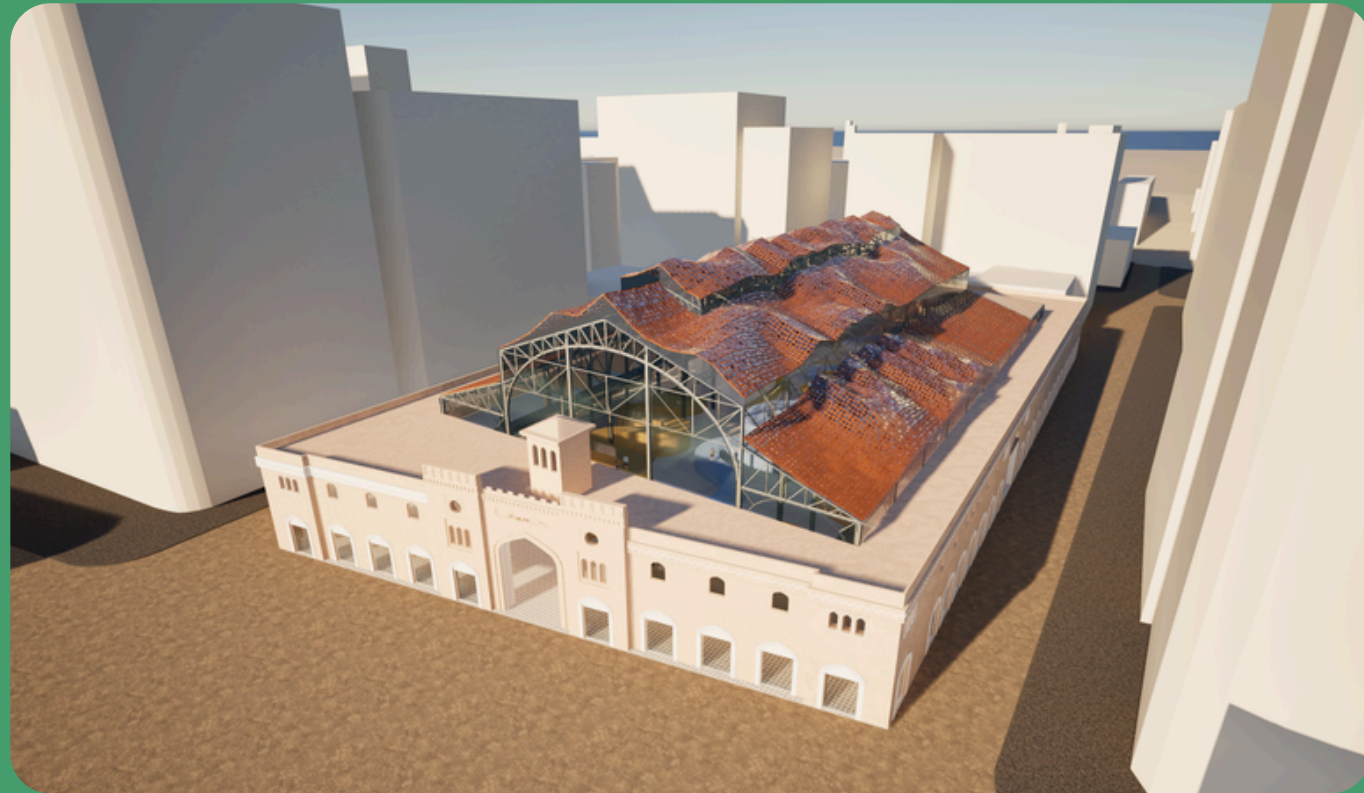


NEVER ENDING CAIRO MARKET

COMPUTATIONAL FORM-FINDING INTERVENTION



TYPE: Complex Forming

YEAR: 2025

TEAM: Andrea Cutroni

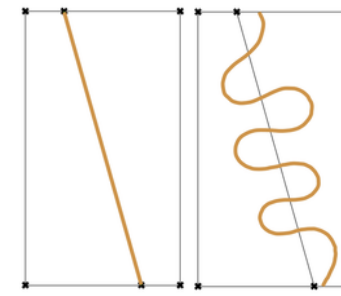
FACULTY: Hesham Shawqy,
Angeliki Maragakis

CONTEXT: IAAC Master in
Advanced Computational Design
(MACAD)

Never Ending Cairo Market revives the historic Bab al-Luq Market in Cairo, Egypt by integrating a redesigned circulation layout and a parametric roof system. Differential growth generates smooth, organic geometries that shape the interior paths and ramps, driving roof frame sizing and deformation. A parametrically reconstructed Cairo tessellation defines the roof modules, producing controlled light-shadow patterns with openings aligned to flows beneath.

DESIGN CONCEPT

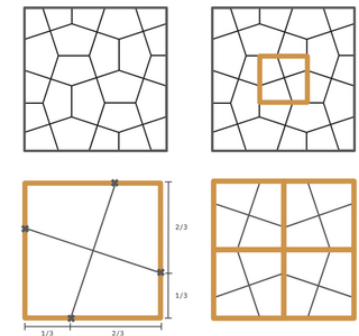
Differential Growth



applied to

Circulation & Roof

Cairo Tessellation



applied to

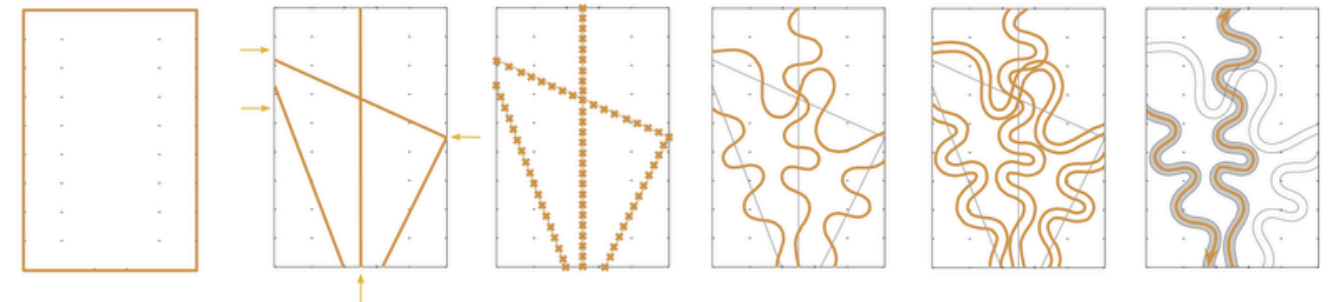
Roof

COMPUTATIONAL DESIGN METHODOLOGY

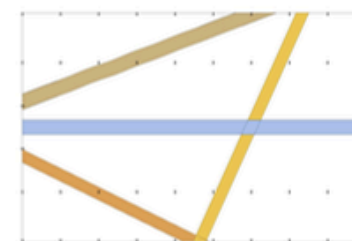
Circulation Layout



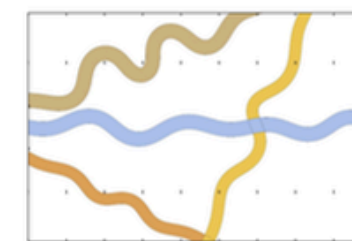
Parameters:
- Sphere Radius
- Length Line



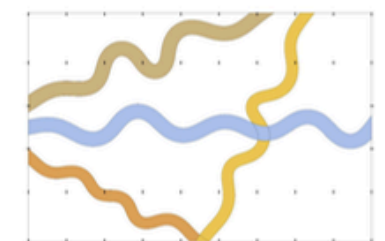
FORM ITERATIONS



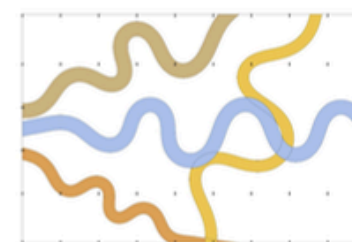
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Length = 0.02 Length = 0.02
Radius = 0.20 Radius = 0.30



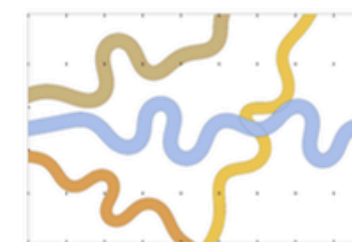
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Radius = 0.40 Radius = 0.45



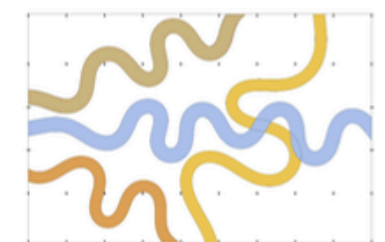
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Radius = 0.30 Radius = 0.45
Length = 0.04 Length = 0.04
Radius = 0.40 Radius = 0.45



Length = 0.02 Length = 0.02
Radius = 0.35 Radius = 0.50
Length = 0.04 Length = 0.04
Radius = 0.60 Radius = 0.60

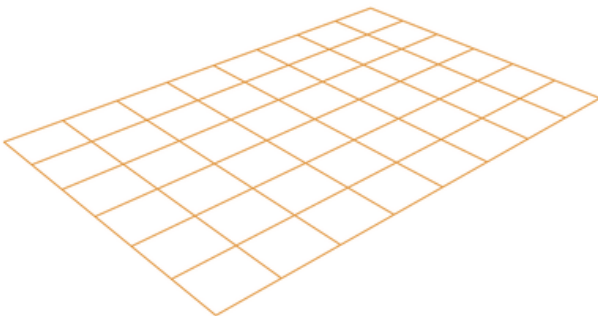


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Radius = 0.40 Radius = 0.60

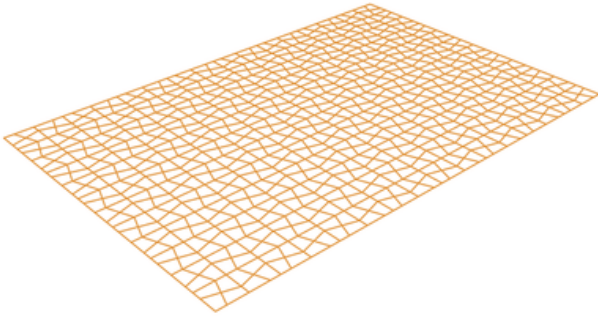


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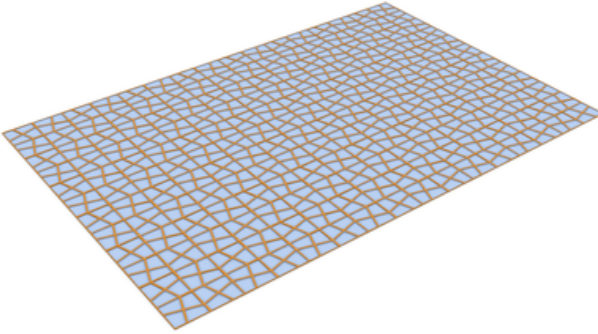
FORM FINDING PROCESS



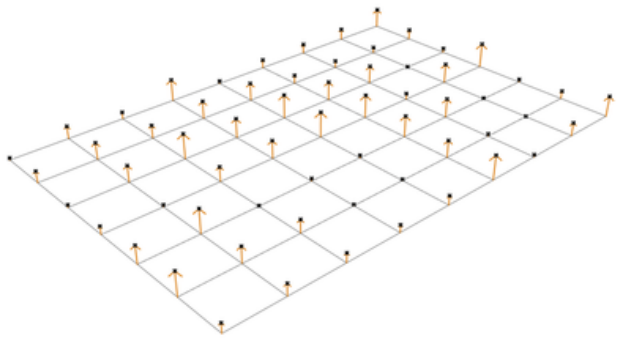
STEP 01
Flat Roof Surface



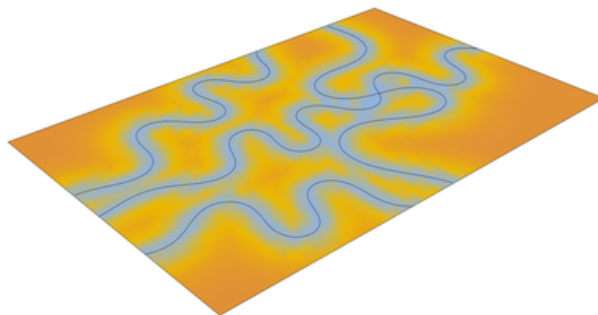
STEP 02
Cairo Tessellation



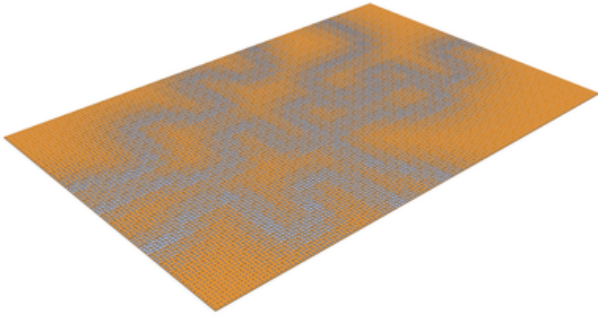
STEP 03
Frame Size Variation



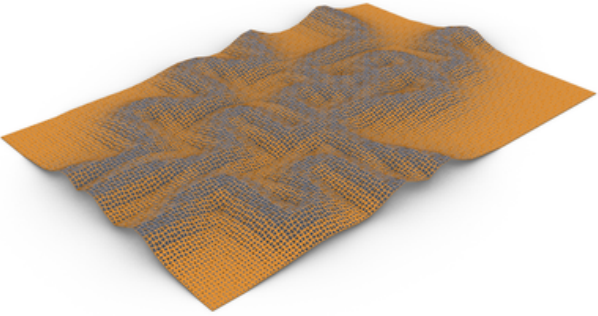
STEP 04
Move Points in Z Direction



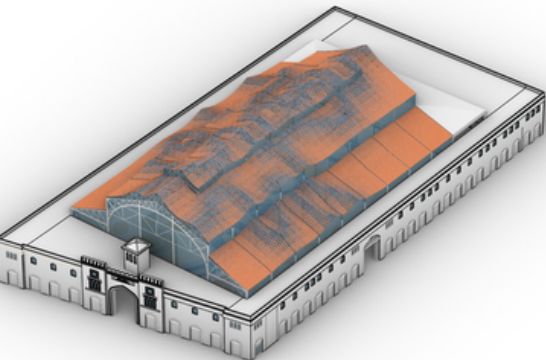
STEP 05
Heat Map based on Circulation Layout



STEP 06
Application to Frame Size



STEP 07
Application to Points Movement



EXPLODED VIEW

